

Computer Science & IT

Database Management System



Query Languages

Lecture No. 03



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Recap of Previous Lecture

Topic

Derived relational algebra operations

- ① Intersection
- ② Join

Topic

Different types of join operations



Topics to be Covered



✓
Topic

Division operation

✱
Topic

Practice questions





Topic : Division ($\div$)

Division operation is used whenever the query is with respect to every or all.

Student (S)

Sid	Sname
S ₁	A
S ₂	A
S ₃	B
S ₄	C

Enroll (E)

Sid	Cid	-----
S ₁	C ₁	
S ₁	C ₂	
S ₂	C ₂	
S ₁	C ₃	
S ₂	C ₃	
S ₃	C ₃	

Course (C)

Cid	Cname
C ₁	OS
C ₂	COA
C ₃	DBMS

Query:- Retrieve Sids of the students who have
Enrolled for all Courses.

Enroll (E)	
Sid	Cid
S ₁	C ₁
S ₁	C ₂
S ₂	C ₂
S ₁	C ₃
S ₂	C ₃
S ₃	C ₃

$\pi_{\text{Sid}} (E) \Rightarrow \text{o/p} =$

 This R.A. Expression
 Will produce the
 Sids of the Students
 who have enroll for
 some Course
 ↓
 { at least }

Sid
S ₁
S ₂
S ₃

Course(C)	
Cid	Cname
C ₁	OS
C ₂	COA
C ₃	DBMS

$\pi_{Cid}(C) \Rightarrow \text{o/p} =$

Cid
C ₁
C ₂
C ₃

This R.A. Expression
will produce
Cids of all
Courses.

* $\pi_{sid}(E) =$

Sid
S ₁
S ₂
S ₃

* $\pi_{cid}(C) =$

Cid
C ₁
C ₂
C ₃

$$\pi_{sid}(E) \times \pi_{cid}(C) \Rightarrow o/p =$$

This R.A. Expression will produce a universal relation, in which Every student who has Enrolled for some Courses is Combined with all Courses.

E.Sid	C.Cid
✓ S ₁	C ₁
✓ S ₁	C ₂
✓ S ₁	C ₃
✓ S ₂	C ₁
✓ S ₂	C ₂
✓ S ₂	C ₃
✓ S ₃	C ₁
✓ S ₃	C ₂
✓ S ₃	C ₃

Student (S)

Sid	Sname
S ₁	A
S ₂	A
S ₃	B
S ₄	C

Enroll (E)

Sid	Cid	fee
S ₁	C ₁	
S ₁	C ₂	
S ₂	C ₂	
S ₁	C ₃	
S ₂	C ₃	
S ₃	C ₃	

Course (C)

Cid	Cname
C ₁	OS
C ₂	COA
C ₃	DBMS

* Retrieve Sids of the students who have Enrolled for all Courses.

* We are actually looking for the Sids in the Enroll table which are associated with all the Cids of the Course table.

* W.r.t. given relational instance o/p should be S₁ because it is the only Sid in Enroll table which is associated with all the Cids of Course table.

W.r.t. "all" we can use division operation,
i.e., $\pi_{Sid, Cid}(E) \div \pi_{Cid}(C)$

Student (S)

Sid	Sname
S ₁	A
S ₂	A
S ₃	B
S ₄	C

Enroll (E)

Sid	Cid	fee
S ₁	C ₁	
S ₁	C ₂	
S ₂	C ₂	
S ₂	C ₃	
S ₃	C ₃	

Course (C)

Cid	Cname
C ₁	OS
C ₂	COA
C ₃	DBMS

* Derivation of division ($\div$) operation using basic relational algebra operation

$$\pi_{\text{Sid}, \text{Cid}}(E) \div \pi_{\text{Cid}}(C)$$

$$\equiv \pi_{\text{Sid}}(E) - \pi_{\text{Sid}} \left[\pi_{\text{Sid}}(E) \times \pi_{\text{Cid}}(C) - \pi_{\text{Sid}, \text{Cid}}(E) \right]$$

Sid
S ₁
S ₂
S ₃

Sid
S ₁

Sid
S ₂
S ₃

Sid	Cid
S ₁	C ₁
S ₁	C ₂
S ₁	C ₃
S ₂	C ₁
S ₂	C ₂
S ₂	C ₃
S ₃	C ₁
S ₃	C ₂
S ₃	C ₃

Sid	Cid
S ₂	C ₁
S ₃	C ₁
S ₃	C ₂

Sid	Cid
S ₁	C ₁
S ₁	C ₂
S ₁	C ₃
S ₂	C ₂
S ₂	C ₃
S ₃	C ₃

Sids of the students who have enroll for some courses

Sids of the students who have enrolled for all courses

In the o/p we will get the Sids of the students who did not enroll for all courses

In the output of this subtraction we will get the Sids of the students who did not enroll for all courses along with the Cids of the course for which they did not enroll.

$$\pi_{sid, cid}(E) \div \pi_{cid}(C)$$

$$\equiv \pi_{sid}(E) - \pi_{sid} \left[\pi_{sid}(E) \times \pi_{cid}(C) - \pi_{sid, cid}(E) \right]$$

①
⑧
⑦
②
④
③
⑥
⑤

Order of
Execution

★ Consider the following relational tables: ✓

★ Supplier (Sid, Sname, Rating)

★ Parts (Pid, Pname, Color)

★ Catalog (Sid, Pid, Cost)

Supplier

<u>Sid</u>	Sname	Rating
S ₁	A	3
S ₂	A	5
S ₃	B	7
S ₄	C	0

Parts

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Q:- Retrieve Sids of
all the suppliers.

π_{sid} (Supplier)

o/p:-

Sid
S ₁
S ₂
S ₃
S ₄

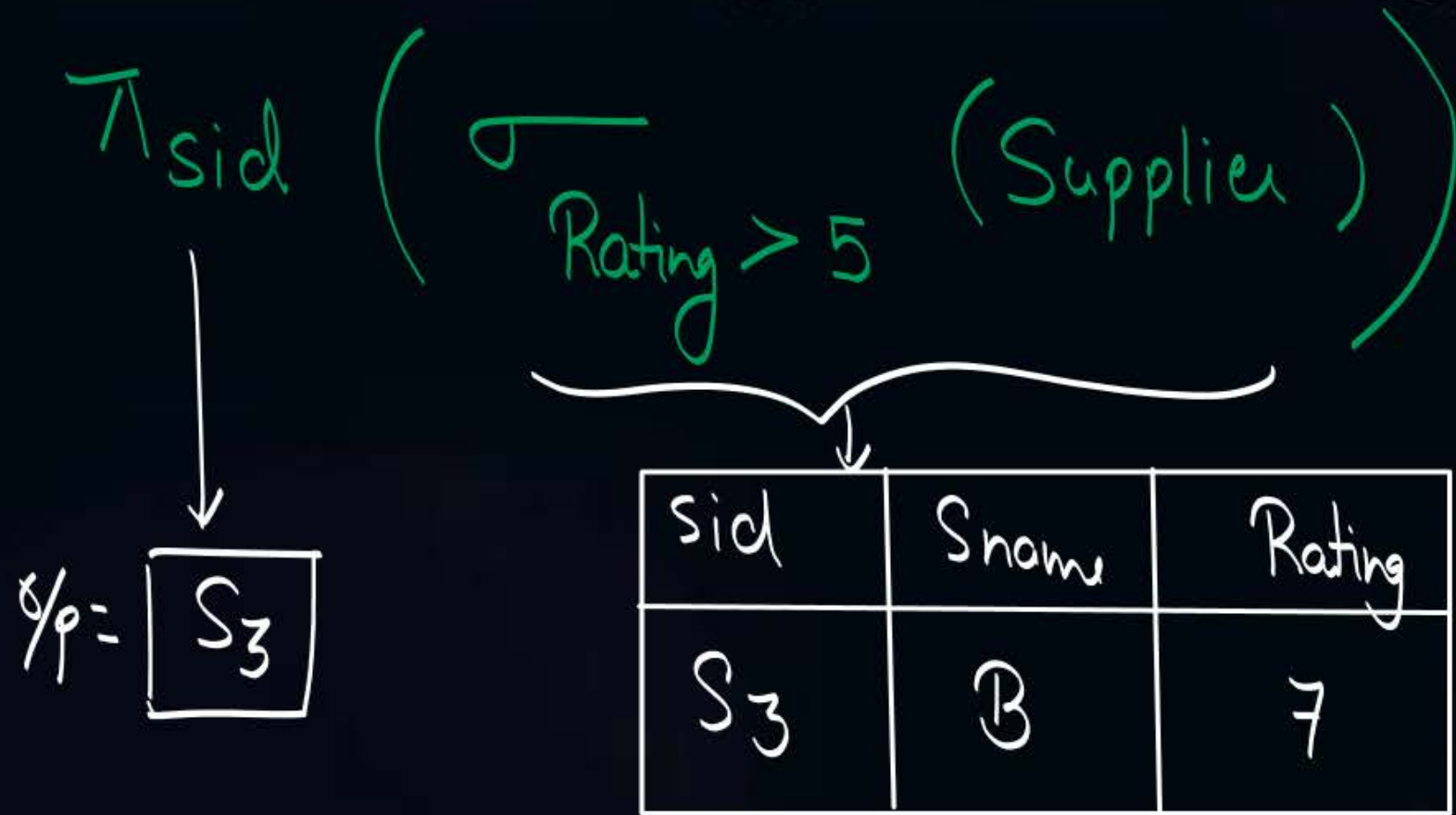
Q:- Retrieve Sids of
Suppliers who have
supplied at least one part.

π_{sid} (Catalog)

o/p =

Sid
S ₁
S ₂
S ₃

#Q. Retrieve $\checkmark$ Sid of the suppliers whose rating is more than 5.



#Q. Retrieve Sid of the suppliers who have supplied some parts.

ie. (at least one)

$\pi_{sid}(\text{Catalog}) \Rightarrow$ o/p :-

Sid
S ₁
S ₂
S ₃



#Q. Retrieve Sid of the suppliers who have supplied some Red color parts.

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Pants

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

(i) $\uparrow$ Catalog.Sid

o/p:-

Sid
S ₁
S ₃

$\uparrow$ Catalog.Pid = Pants.Pid (Catalog x Pants)

$\uparrow$ Pants.Color = 'Red'

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

(ii) $\nearrow$ Sid

$\downarrow$
%p =

S ₁
S ₃

$\left[\sigma_{\text{Color} = 'Red'} (\text{Catalog} \bowtie \text{Parts}) \right]$

Parts

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S ₁	P ₁	10	P ₁	XYZ	Red
S ₁	P ₁	10	P ₂	PQR	Green
S ₁	P ₁	10	P ₃	XYZ	Red
S ₂	P ₂	20	P ₁	XYZ	Red
S ₂	P ₂	20	P ₂	PQR	Green
S ₂	P ₂	20	P ₃	XYZ	Red
S ₃	P ₃	30	P ₁	XYZ	Red
S ₃	P ₃	30	P ₂	PQR	Green
S ₃	P ₃	30	P ₃	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S ₁	P ₁	10
S ₂	P ₂	20
S ₃	P ₃	30

Parts

<u>Pid</u>	Pname	Color
P ₁	XYZ	Red
P ₂	PQR	Green
P ₃	XYZ	Red

$$(iii) \pi_{Catalog.Sid} \left(\sigma_{Catalog.Pid = Parts.Pid} \left(Catalog \times \left(\sigma_{Color = 'Red'} (Parts) \right) \right) \right)$$

$$(iv) \pi_{sid} \left(Catalog \bowtie \left(\sigma_{Color = 'Red'} (Parts) \right) \right)$$

These two query are more efficient
Compared to previous two
queries.

HW:

#Q.

Retrieve Sid of the suppliers who have supplied some Red or some
Green color parts.



H.W.



#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

H.W.



#Q. Retrieve Sid of the suppliers who have supplied all parts.

H.W.



#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

H.W.

#Q.

Retrieve Sid of the suppliers who have supplied at least two parts.



HW:

#Q.

Retrieve Sid of the suppliers who have supplied exactly one part.



H.W.

#Q.

Retrieve Sid of the suppliers who have supplied at most one part.



H.W.

#Q.

Retrieve Sid of the suppliers who have supplied at least three parts.



H.W.

#Q.

Retrieve Sid of the suppliers who have supplied most expensive parts.





2 mins Summary



✓
Topic

Division operation

✓
Topic

Practice questions

A pink checkmark is positioned above the word 'THANK' in the text.

THANK - YOU